

Airline Data Management and Analysis Using Power BI

Problem Statement

The airline industry operates with numerous complexities, requiring effective data management and insights into flight schedules, passenger details, and ticketing systems. This project aims to analyze airline operations for improving efficiency and customer satisfaction by leveraging Power BI's data visualization and analytical features.

Datasets Used

Flight Information: Includes FlightID, FlightNumber, Airline, Destination, and Status.

Passenger Information: Includes PassengerID, FlightID, and SeatNumber.

Ticket Information: Includes TicketID, FlightID, and BookingStatus.

Objective

To analyze and visualize airline data for operational insights, passenger management, and ticket booking trends using Power BI.

Tasks and Deliverables

Task 1: Data Preparation and Cleaning

- Extracted data and Performed in Power Query Editor.
- Removed duplicate
- Removed empty cells
- Changed data types appropriately

FlightID	FlightNumber	Airline	Destination	Status
1	1001 FL1102	Airline D	Houston	On Time
2	1002 FL1435	Airline B	Chicago	On Time
3	1003 FL1860	Airline A	New York	Cancelled
4	1004 FL1270	Airline C	Chicago	Delayed
5	1005 FL1106	Airline C	New York	Delayed
6	1006 FL1071	Airline A	Phoenix	On Time
7	1007 FL1700	Airline C	Los Angeles	Cancelled
8	1008 FL1020	Airline C	Los Angeles	Delayed
9	1009 FL1614	Airline A	Los Angeles	Cancelled
10	1010 FL1121	Airline D	Chicago	Cancelled
11	1011 FL1466	Airline A	Phoenix	On Time
12	1012 FL1214	Airline D	New York	Delayed
13	1013 FL1330	Airline C	Houston	On Time
14	1014 FL1458	Airline C	New York	Delayed

flight_information	123	PassengerID	123	FlightID	A6	SeatNumber
passenger_information	1		2		1161	38A
ticket_information	2		3		1157	24D
	3		4		1141	30B
	4		5		1046	17E
	5		6		1035	29D
	6		7		1134	10A
	7		8		1082	10A
	8		9		1115	20E
	9		10		1197	34E
	10		11		1047	2E
	11		12		1153	43C
	12		13		1194	48C
	13				1010	47A

Task 2: Data Modeling

- Load all 3 tables into Power BI.
- Go to the Model View.
- Create relationships using FlightID:
 - FlightID in Flight linked to FlightID in Passenger
 - FlightID in Flight linked to FlightID in Ticket
- Ensure correctly matched (One-to-Many from Flight to others).

← Edit relationship



Select tables and columns that are related.

From table

passenger_information

FlightID	PassengerID	SeatNumber
1161	1	38A
1157	2	24D
1141	3	30B

To table

flight_information

Airline	Destination	FlightID	FlightNumber	Status
Airline D	Houston	1001	FL1102	On Time
Airline B	Chicago	1002	FL1435	On Time
Airline A	Phoenix	1006	FL1071	On Time

Cardinality

Many to one (*:1)

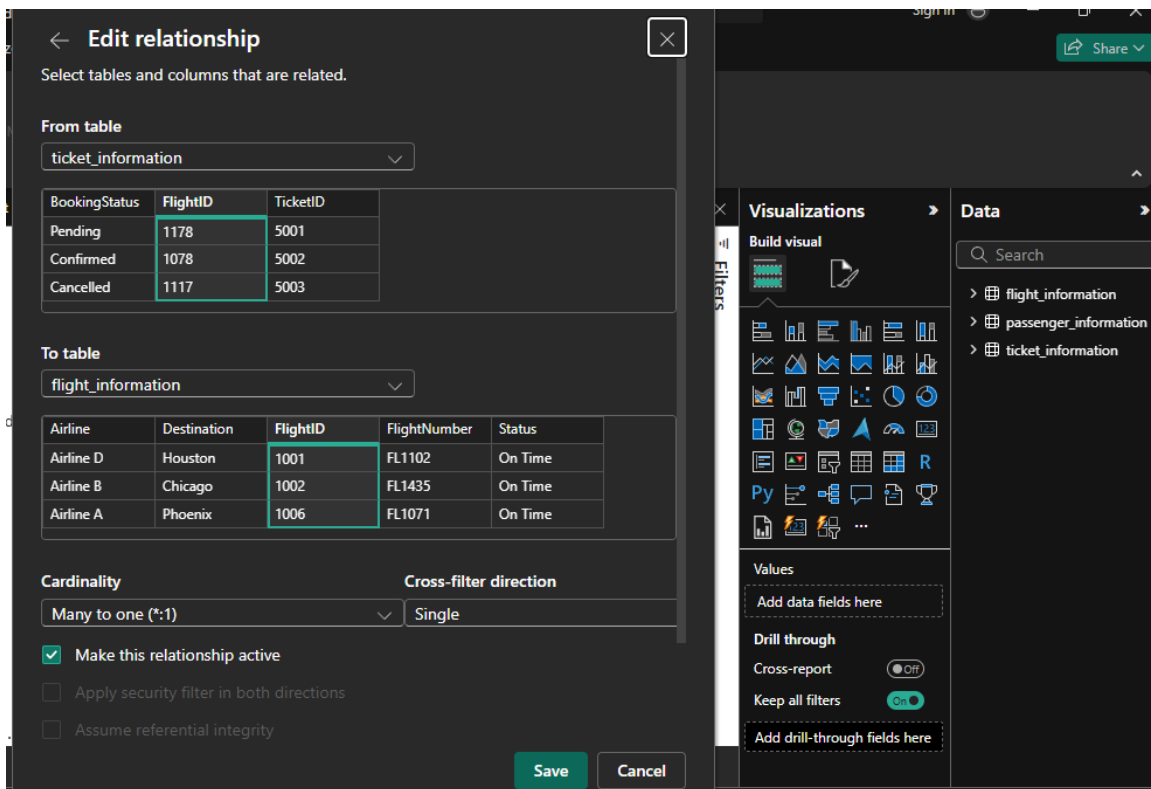
Cross-filter direction

Both

- ☒ Make this relationship active
- ☐ Apply security filter in both directions
- ☐ Assume referential integrity

Save

Cancel



Task 3: Enhanced Data Insights

- Add a **Conditional Column**: If Status = "On Time", then "Best", else "To Be Improved".
- Use **Column from Examples** to extract just the Flight Number from a field like "FL1102 to 1102".

Queries [3]

flight_information

passenger_information

ticket_information

fx

Table.AddColumn(#"Removed Columns", "Classified_flights", each if Text.Contains([Status], "On Time") then "Best"

	FlightNumber	Airline	Destination	Status	Classified_Rights
1	1001	Airline D	Houston	On Time	Best
2	1002	Airline B	Chicago	On Time	Best
3	1003	Airline A	New York	Cancelled	To Be Improved
4	1004	Airline C	Chicago	Delayed	To Be Improved
5	1005	Airline C	New York	Delayed	To Be Improved
6	1006	Airline A	Phoenix	On Time	Best
7	1007	Airline C	Los Angeles	Cancelled	To Be Improved
8	1008	Airline C	Los Angeles	Delayed	To Be Improved
9	1009	Airline A	Los Angeles	Cancelled	To Be Improved
10	1010	Airline D	Chicago	Cancelled	To Be Improved
11	1011	Airline A	Phoenix	On Time	Best
12	1012	Airline D	New York	Delayed	To Be Improved
13	1013	Airline C	Houston	On Time	Best
14	1014	Airline C	New York	Delayed	To Be Improved
15	1015	Airline C	Houston	Delayed	To Be Improved
16	1016	Airline B	New York	Delayed	To Be Improved

Query Settings

PROPERTIES

Name
flight_information

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Filtered Rows
- Removed Duplicates
- Removed Columns
- Added Conditional Column

flight_information

passenger_information

ticket_information

Enter sample values to create a new column (Ctrl+Enter to apply).

Transform: TextAfterDelimiter([FlightNumber], 'L')

OK Cancel

	FlightID	FlightNumber	Airline	Destination	flightnumber
1	1001	FL1102	Airline D	Houston	1102
2	1002	FL1435	Airline B	Chicago	1435
3	1003	FL1860	Airline A	New York	1860
4	1004	FL1270	Airline C	Chicago	1270
5	1005	FL1106	Airline C	New York	1106
6	1006	FL1071	Airline A	Phoenix	1071
7	1007	FL1700	Airline C	Los Angeles	1700
8	1008	FL1020	Airline C	Los Angeles	1020
9	1009	FL1614	Airline A	Los Angeles	1614
10	1010	FL1121	Airline D	Chicago	1121
11	1011	FL1466	Airline A	Phoenix	1466
12	1012	FL1214	Airline D	New York	1214
13	1013	FL1330	Airline C	Houston	1330

PROPERTIES

Name

flight_information

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Filtered Rows

Removed Duplicates

Removed Columns

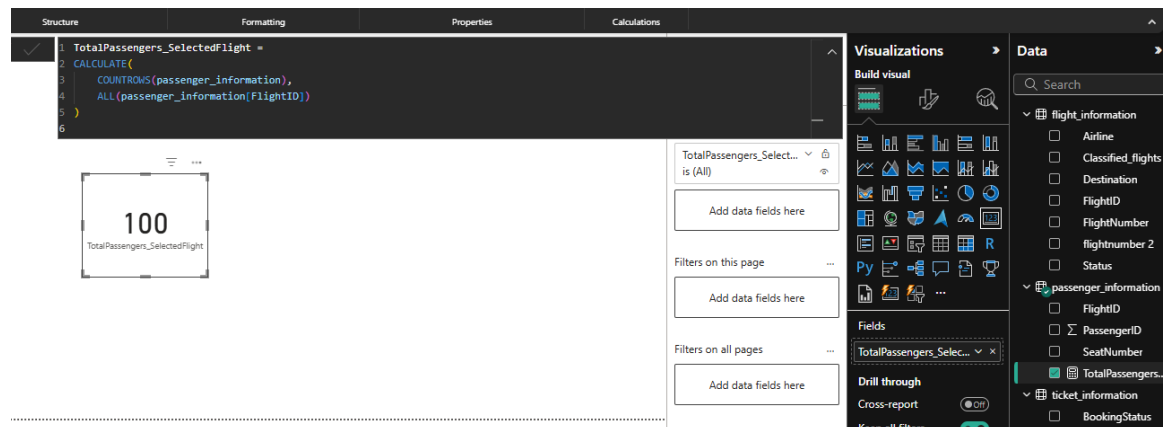
Added Conditional Column

Column From Examples

Task 4: Calculations Using DAX

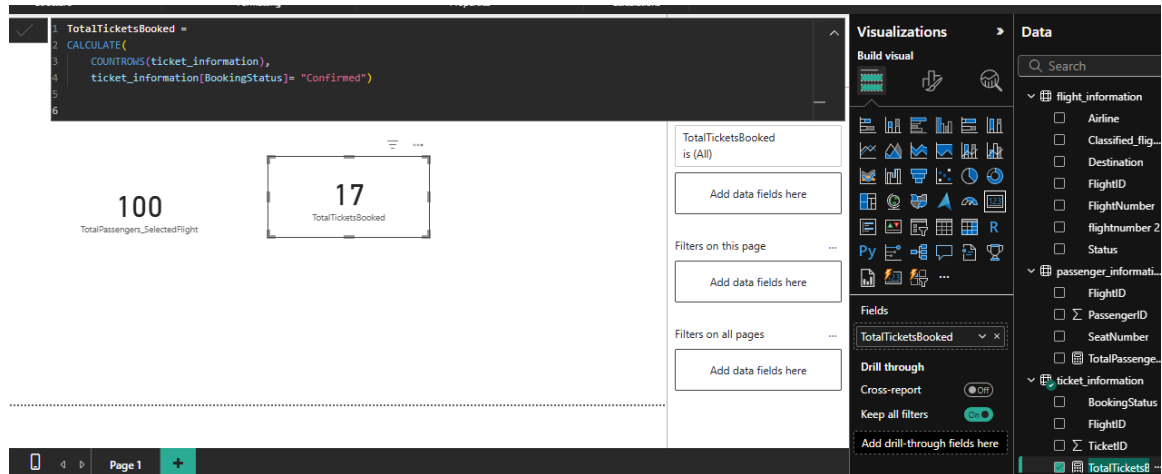
- DAX Calculations:
- Total Passengers for a Selected Flight:

```
TotalPassengers_SelectedFlight = CALCULATE ( COUNTROWS
(Passenger Information)ALL(Passenger Information [FlightID]))
```



- Total Tickets Booked:

TotalTicketsBooked =
CALCULATE(COUNTROWS(Ticket_information),
Ticket_information[BookingStatus] = "Confirmed")



- Filtered Table of "Best" Flights only:

Here we can you use either table view or table chart and go to filter pane change “Classified_Flights” to “Best”.

The screenshot shows a Power BI table view with the following columns: FlightID, Destination, FlightNumber, and Classified_flights. The table contains 20 rows of data, all of which are classified as 'Best'. The 'Filters' pane on the right shows 'Classified_flights' set to 'Best', with a count of 82. The 'Visualizations' pane on the right shows a table icon selected, and the 'Data' pane shows the 'flight_information' table with fields like 'Airline', 'Classified_f...', 'Destination', 'FlightID', 'FlightNumber', 'flightnumber 2', and 'Status'.

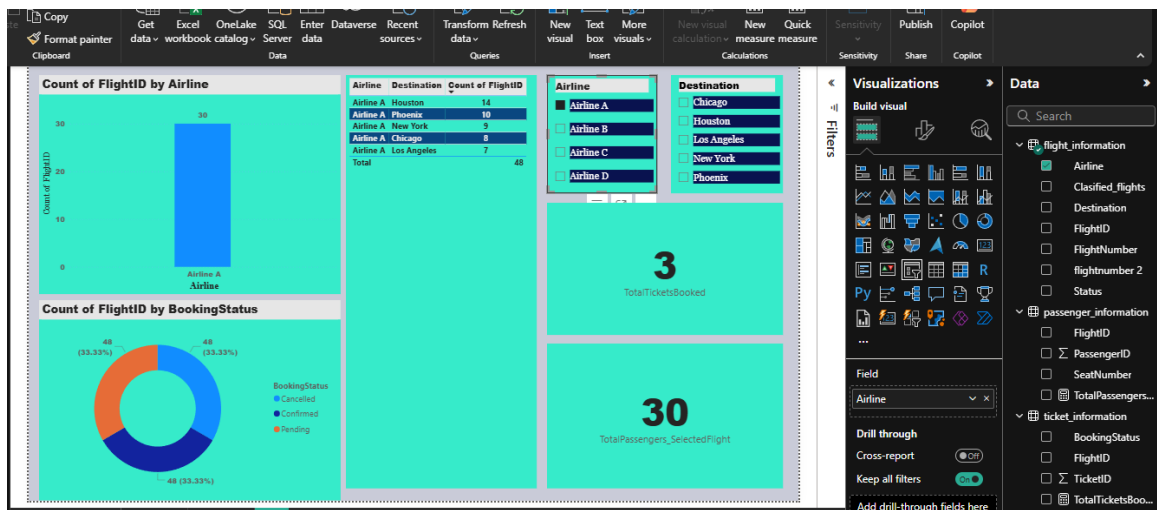
FlightID	Destination	FlightNumber	Classified_flights
1001	Houston	FL1102	Best
1002	Chicago	FL1435	Best
1006	Phoenix	FL1071	Best
1011	Phoenix	FL1466	Best
1013	Houston	FL1330	Best
1020	New York	FL1130	Best
1023	Chicago	FL1769	Best
1025	Phoenix	FL1491	Best
1027	Chicago	FL1805	Best
1028	Chicago	FL1385	Best
1029	Los Angeles	FL1191	Best
1030	Phoenix	FL1955	Best
1031	New York	FL1276	Best
1033	New York	FL1459	Best
1034	Phoenix	FL1313	Best
1036	Phoenix	FL1252	Best
1039	Chicago	FL1560	Best
1043	Houston	FL1681	Best
1044	Phoenix	FL1475	Best
1046	Chicago	FL1975	Best
1048	New York	FL1189	Best
1050	Phoenix	FL1686	Best

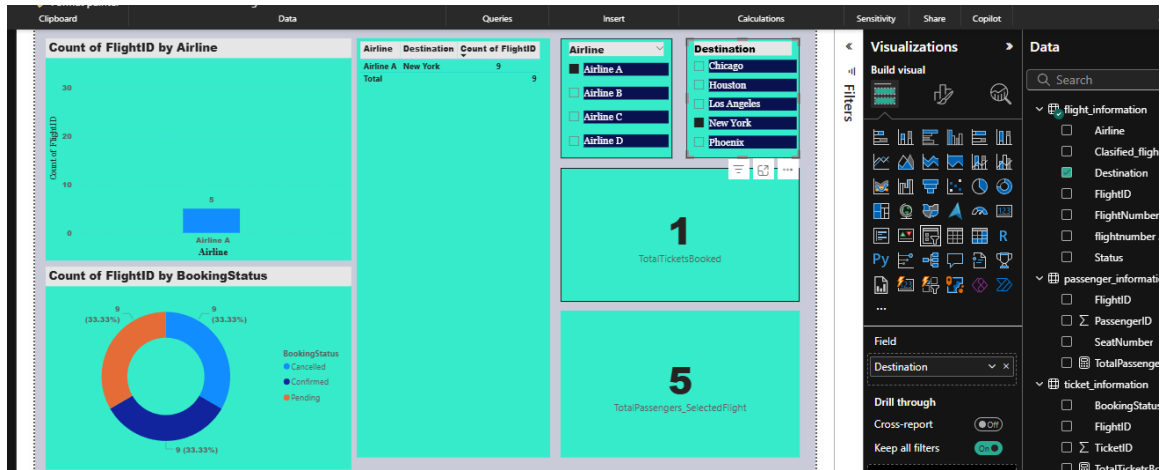
Task 5: Visualizations and Interactive Features

- Use BAR CHART to see the report of number of Passengers by Airline
X-Axis = “Airline” from Flight information table
Y-Axis= “Count of Flight” from Passengers information table
- Use DONUT CHART to see the report “% of Flights by Booking status”
Legends = “Booking Status” from Ticket information
Values= “Count of FlightID” from Flight information
- Use Table view report for “Flights by Airline and Destination

Column = Airline, Destination, Count of FlightID

- Interactive Features:
 - Slicers for Airline and Destination
 - Cards for quick views of “TOTAL TICKETS BOOKED” and “TOTAL PASSENGERS FOR SELECT FLIGHT”

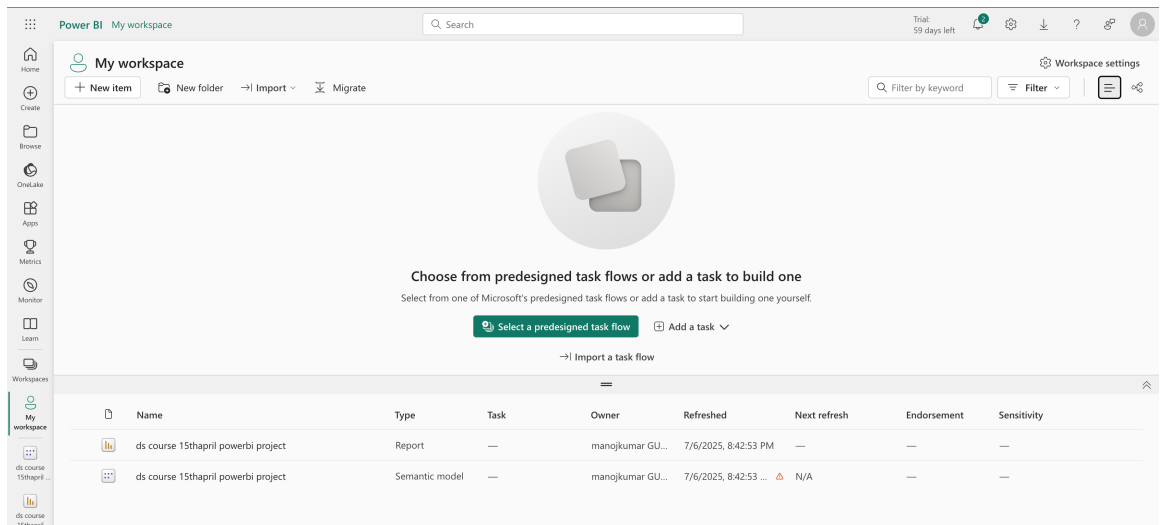
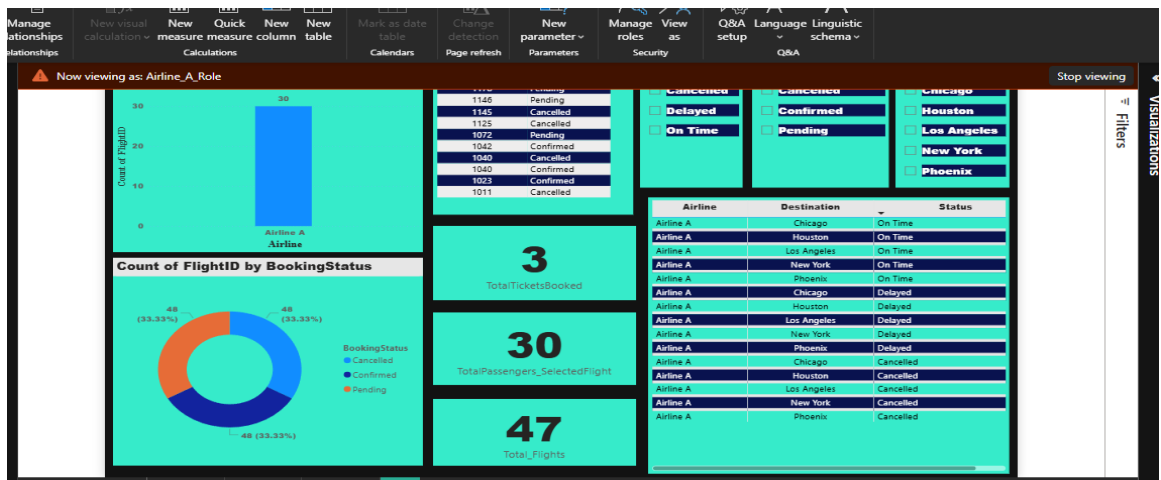
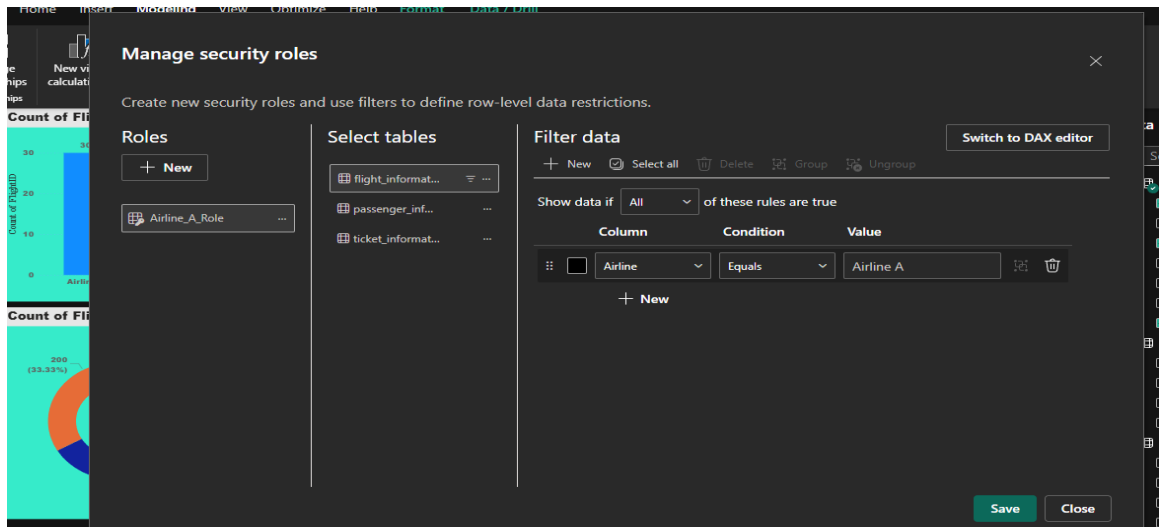


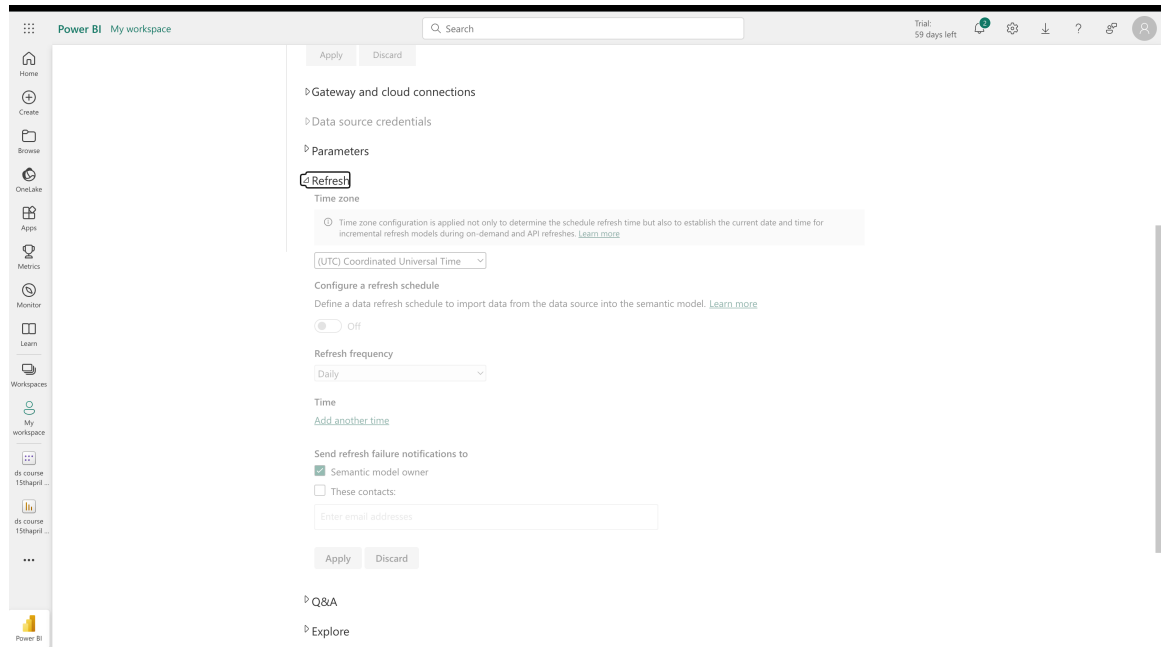


Task 6: Final Dashboard and Power BI Service

- Created a clean dashboard with slicers, Cards, and Table charts.
 - Use slicers for “Flight Status”, “Booking Status” and “Destination”.
 - Use “Total Ticked Booked”, “Total Passengers” and “Total Flights” Cards for Quick View.
 - Use Table Charts to see which flight, Destination, Airline and all Statuses.
- Row-Level Security (RLS): In the Modeling, click on manage roles then Create Role created: AirlineA_Role
Filter used: [Airline] = "Airline A"
- Assigned to user via Power BI Service.
- Scheduled Refresh: Set to 5:00 PM daily.







Key Insights

- Airline A had the highest number of passengers and moderate Cancelled Flights.
- Airline B had a least number of passengers rate and Least number of C
- Phoenix State had the highest volume Flights.
- There was no flights to New York is “On Time”.

EXPLANATION :

<https://drive.google.com/file/d/1cyvkkIZeJ9D4wahq00LICVfFCIXkSO1M/view?usp=sharing>